

12-16 B Cu)

Mol Balance For CSTR is

$$V = \frac{F_{A0} X}{-r_A}$$

$$F_{A0} = (V_0)(C_{A0})$$

For CSTR : this case

~~WAAAAA~~

$$K \left(C_A - \frac{C_B}{K_L} \right) = -r_A$$

so...

$$= \frac{V_0 C_{A0} X}{K \left(C_A - \frac{C_B}{K_L} \right)}$$

$$C_B = \cancel{C_{A0}(1-X)} C_{A0} X$$

$$C_A = C_{A0}(1-X)$$

$$V = \frac{V_0 C_{A0} X}{K \left(C_{A0} \left((1-X) - \frac{X}{K_L} \right) \right)}$$

$$\frac{V}{V_0} = \frac{X}{K \left((1-X) - \frac{X}{K_L} \right)}$$

$$K \frac{V}{V_0} = \frac{X}{\left((1-X) - \frac{X}{K_L} \right)}$$

$$\rightarrow \frac{V}{V_0} = \tau$$

$$\therefore K \tau = \frac{X}{(1-X) - X/K_L}$$

$$X = K \tau \left((1-X) - X/K_L \right)$$

~~WAAAAA~~

After Throwing into solve to isolate x I get

$$X = \frac{\tau K}{1 + \tau K \left(1 + \frac{1}{K_L} \right)}$$

Let's plug in now

Given

$$\ln(T) = (-\Delta H_{rxn})(X)$$

$$-\Delta H_{rxn} = 80000$$

$$K = 1 \text{ min}^{-1}$$

$$\tau = \frac{V}{V_0} = \frac{10}{1} = 10 \text{ min}$$

$$K_L = 100$$

$\therefore$

$$X = \frac{(10)(1)}{1 + (10)(1) \left(1 + \frac{1}{100} \right)}$$

$$X = \frac{10}{11.1} = 0.9001$$

$$\therefore \ln(T) = (80000)(0.9001) 72,072 \text{ Cal/mol}$$

b) @ Points 376k & 419k look at Mat lab plot. Also 310 k.
Code varies slightly from book answer but it is very close.

c) Look at graph. Answer is 310 & 419k. Since at these points
 $R > G \rightarrow$

d) $R_T = C_{PA} (1 + K) (T - T_C)$

$$K = \frac{U_A}{F_{AO} \cdot C_{PA}}$$

$$R_T = C_{PA} \left(1 + \frac{U_A}{(F_{AO})(C_{PA})} \right) (T - T_C)$$

$$R_T = C_{PA} \left(1 + \frac{U_A}{F_{AO}} \right) (T - T_C)$$

$$T_C = \frac{T_0 + K T_A}{1 + K} \Rightarrow \frac{310 + (9)(310)}{1 + 9} \Rightarrow \frac{310 + 2790}{10} \Rightarrow \underline{310}$$

For simplicity

$$K = \frac{3600}{(10)(40)} = \frac{3600}{400} = 9 \quad \uparrow$$

$$\therefore R_T = 40 \left(1 + \frac{3600}{10} \right) (T - 310) \Rightarrow 400(T - 310) \quad T = 419$$

$$R_T = 400(419 - 310) \Rightarrow 43600$$

$$R_T = G_T$$

$$G_T = (80000)(X) = 43600$$

$$\boxed{X = 0.545}$$

E) I could not get a good plot ~

F) Look @ graph on MatLab. When $U_A = 0$ Max $T = 431 \text{ K}$

G) Want Max U_A . From plot the highest point $\approx 405 \text{ K}$ (GT plot);
GT @ that point ≈ 733574 .

$$GT = RT \Rightarrow 733574 = C_{PA} (1 + K) (T - T_c)$$

$$733574 = C_{PA} \left(1 + \frac{U_A}{F_{A0} C_{PA}} \right) (T - T_c)$$

$$733574 = \left(C_{PA} + \frac{U_A}{F_{A0}} \right) (T - T_c)$$

$$733574 = \left(40 + \frac{U_A}{10} \right) (405 - 310) \frac{310 + \left(\frac{U_A}{F_{A0} C_{PA}} \right) (310)}{1 + \frac{U_A}{F_{A0} C_{PA}}}$$

$$733574 = \left(40 + \frac{U_A}{10} \right) (405 - 310) + \frac{0.775 U_A}{1 + \frac{U_A}{400}}$$

using online solver $U_A = 76818$

h) What if we were working with a PFR instead of a CSTR.

This is critical thinking because it requires us to change perspective.

i) I have no idea.

j) Operate @ lower T_A